

# A statistical scaling model for fracture network geometry, with validation on a multiscale mapping of a joint network (Hornelen Basin, Norway)

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[1] Fracture patterns are characterized by a complex geometry which involves a large length distribution and nonhomogeneous density distributions. Here we address the issue of the modeling of this complex geometry over scales through a first-order model providing the characterization of the number of fractures of a given length and at a given scale. We propose that the simplest model is a power law in both space and fracture length, with three main parameters: the exponent  $a$  of a power law length distribution, the fractal dimension  $D$  which fixes the scale dependence of the number of fractures, and the fracture density  $\alpha$ . We verify the applicability of this model on seven fracture patterns mapped from the metric scale up to almost the kilometric scale in the Hornelen basin. The model efficiently describes fracture network properties at all scales with a single set of parameters. For the Hornelen fracture networks we found a simple relationship between basic exponents,  $a = D + 1$ , implying that the fracture network is self-similar. Through this analysis, we present different methodological developments for deriving the basic exponents  $a$  and  $D$  and for verifying the consistency of the model. Overall, the main methodological development is about the normalization of the measurements made from different scales of observation. Finally, we discuss both the limitations and the uses of such a model for analyzing the hydraulic and mechanical properties of fracture

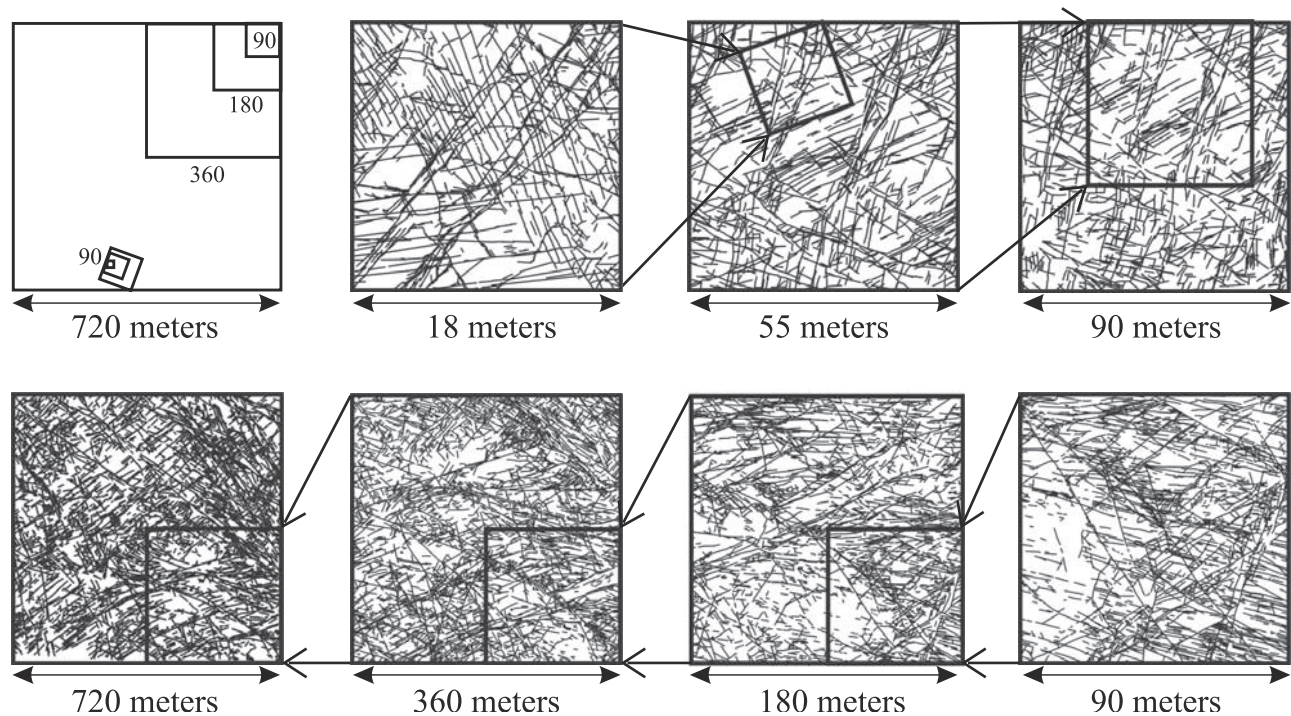
networks. **INDEX TERMS:** 3250 Mathematical Geophysics: Fractals and multifractals; 5104 Physical Properties of Rocks: Fracture and flow; 8010 Structural Geology: Fractures and faults; **KEYWORDS:** fractures, faults, fractal, scaling laws, length distribution, geometry

## 1. Introduction

[2] Fractal concepts applied to fracture networks predict a generic relationship between fractures at all scales [Allègre *et al.*, 1982; King, 1983; Turcotte, 1986]. For the last 15 years, such a scaling concept has been largely debated from statistical studies of detailed fracture maps [see Bonnet *et al.*, 2001, and references therein]. A large number of works analyzed either the spatial distribution of fractures [Barton and Larsen, 1985; Okubo and Aki, 1987; LaPointe, 1988; Hirata, 1989; Gautier and Lake, 1993; Ouillon *et al.*, 1996a] or their length distribution [Segall and Pollard, 1983; Gudmunsson, 1987; Villemain and Sunwoo, 1987; Childs *et al.*, 1990; Main *et al.*, 1990; Scholz and Cowie, 1990; Davy, 1993; Odling, 1997] to test the power law scaling model.

[3] Despite these numerous analyses, the scientific community is still largely debating the statistical relevance of power law models, mainly because of the difficulty of

achieving a robust statistical analysis on limited data sets [Pickering *et al.*, 1995; Clark *et al.*, 1999; Bonnet *et al.*, 2001]. In general, scaling laws are obtained over only 1 order of magnitude, which is clearly insufficient to demonstrate the fractal nature of fracture patterns [Feder, 1988]. Moreover, the methods used to achieve a statistical analysis are sometimes subject to discussion [see Bonnet *et al.*, 2001], and not all fracture patterns are characterized by a fractal geometry [Odling *et al.*, 1999]. More generally, there is currently no statistical model that globally describes the geometry of fracture networks, including the density term, the length and orientation distribution, spatial correlation, etc. Generally, fractal dimension or power law length exponent is determined independently, and the general statistical model that should at least involve these two scaling parameters is not tested. There is clearly a need for determining a sound statistical model, at least for relating fracture geometry to mechanical or transport properties. Theoretical studies that investigate relationships between geometry and physical properties [Ross, 1986; Turcotte, 1986; Marrett and Allmendinger, 1991; Davy *et al.*, 1990; Acuna and Yortsos, 1995; Bour and Davy,



**Figure 1.** Maps of the seven fracture patterns from the Hornelen basin [Odling, 1997]. The procedure used to map the different fracture patterns is described by Odling [1997].

1997, 1998; Berkowitz *et al.*, 2000] use existing rudimentary geometrical models and clearly do not take into account the bulk complexity of fracture network geometry provided by structural geologists.

[4] Here we address the issue of modeling this complex geometry over a range of scales through a first-order statistical model. It consists first in qualifying a fracture with a number of properties such as its length,  $l$ , its orientation  $\theta$ , its throw,  $t$ , etc. The first-order model gives the number of fractures having a length  $l$ , an orientation  $\theta$ , etc., contained in an elementary volume of characteristic scale  $L$ . Second-order statistics would involve the spatial correlation between the different fracture parameters. An example of such correlation was given by Bour and Davy [1999], who showed statistically that small faults are more clustered than large ones. This is typically a second-order statistical property that relates local fault density and fault length.

[5] We restrict our analysis to a first-order model that yields the number of fractures of a given length and at a given scale. We propose that the simplest first-order model involves at least three parameters: the exponent  $a$  of a power law length distribution, the fractal dimension  $D$  which fixes the scale dependence of the number of fractures, and the fracture density  $\alpha$ . We verify the applicability of this model on seven two-dimensional fracture patterns (Figure 1) from the metric scale up to almost the kilometric scale in the Hornelen basin.

## 2. Hornelen Data Set

[6] The data set we used to test the model consists of seven two-dimensional fracture patterns mapped in the same area from the metric scale up to almost the kilometric scale (Figure 1) [Odling, 1997]. Each map contains more than

2000 fractures (Table 1). The area from which the seven fracture maps were made contains a system of joints (tension fractures) in well-indurated Devonian age sandstones of the Hornelen basin in western Norway, some 500 km north of Bergen. The rocks are composed of sandstones which are organized first into beds of the order of 2 m thick and second into cycles some 100–200 m thick. The joints are mostly close to perpendicular to the sedimentary layering, and the geometry of the joint system is easily seen, because of excellent exposure, both on the tops of sedimentary cycles and in cliff faces. Studies of this joint system [Odling, 1997] show that at outcrop scale, single traces represent single joints but at kilometer scale, traces represent joint zones that are complex structures of well-connected densely packed joints. The width of these joint zones correlates with trace length [Odling, 2001] and with their visual signature. The seven maps (Figure 1) were made from outcrop observations and aerial photographs at a range of resolutions. Thus each map represents a limited scale range of joint trace lengths controlled by the resolution of the image. It was found that the joints are vertically

**Table 1.** Values of  $N(L)$  and  $M(L)$ <sup>a</sup>

| $L$ | $N(L)$ | $M(L)$ |
|-----|--------|--------|
| 18  | 2,123  | 1,198  |
| 55  | 2,565  | 4,720  |
| 90  | 3,429  | 8,218  |
| 90  | 2,440  | 7,797  |
| 180 | 3,137  | 17,823 |
| 360 | 3,992  | 40,043 |
| 720 | 4,092  | 80,325 |

<sup>a</sup> $N(L)$ , the number of fractures included in the system, and  $M(L)$ , the cumulative length of fracture included in the system (in meters) for the different fracture networks ( $L$  is the system size, in meters).

persistent with aspect ratios close to 1.0. Thus, in this case, the sedimentary layering has had only a minor influence on the development of the joint system which can be regarded as of “nonstratabound” type [Odling *et al.*, 1999], i.e., close to a fracture system developed in a mechanically isotropic rock mass. Such systems often display power law fracture length distribution and a tendency to spatial distribution that is more clustered than a spatially random distribution [Odling *et al.*, 1999]. Thus this fracture system may be thought of as typical in style to joints systems in massive rocks, systems that are wide-spread in nature.

### 3. First-Order Model of Fracture Length and Density

#### 3.1. Fractured System to Which the Statistical Model Applies

[7] The statistical properties that we are looking for require a clear definition of what we consider to be the fractured system. A clear and unambiguous definition of fractured system is the space, empty in general, created during the fracturing process. Nobody measures the fracture volume because fracture thickness is most of the times smaller than observation scale. Rather, geologists define the fractured system as material surfaces that wrap the fractured volume. This definition is physically sound since most of fracturing theories are based on the energy necessary to create such surfaces. It is not totally unambiguous geometrically speaking because these surfaces have a small-scale complexity (due to an intrinsic self-affine property or to the formation of a gouge) that makes their measure scale-dependent if the resolution scale is very small (which is barely reached in practice). Classical box-counting methods or mass dimension were actually applied on this fractured-surface system, or on the one-dimensional (1-D) trace networks defined by its intersection with the outcrop plane. Geologists generally go further and consider the fractured system as a population of fractures that are individual elements, roughly planar, in which the displacement is supposed to be spatially coherent, that is, predictable from place to place. A basic property such as fracture length, or fracture orientation, does make sense only if each fracture is individualized, which is not the case for the fractured-surface system defined above. This decomposition into individual elements is coherent with the physics of cracks, in which the stress and strain field due to the crack loading is related to the distance to crack tips, and to the crack surface.

[8] The fracture population system or the fractured-surface system is obviously referring to the same fractured system; the differences lie only in the descriptors. The fracture population system is characterized by a fracture density (number of fractures encountered in a material volume), a fracture length distribution, a fracture orientation distribution, fracture-to-fracture distances; the fracture surface system is characterized by a surface density that is the total fracture surface in a material volume (whose unit is  $\text{m}^{-1}$  whatever the space dimension). The equivalence between these properties is not trivial: The surface fracture density is a function of the individual fracture density (that is a first-order descriptor of this population) and of potential correlations between fracture length, position or orientation [Bour, 1997; Bour and Davy, 1999]. In this paper, we focus

on the first-order model of the fracture population system, that is, the density of fractures having a length  $l$ . In a sense, this is the most basic descriptor of the fractured system because it does not contain any information on correlation between fracture properties. However, it gives some basic but meaningful scaling properties on the fracture position correlation and on the length distribution. Note that we already found different scaling regimes between the fractured-surface system and the fracture population system [Bour, 1997]. The fractured-surface network has a trivial dimension of  $\sim 2$  [Odling, 1992; Bour, 1997], while the fractal dimension of fracture centers is  $\sim 1.8$  (see below). We will address this issue in a future paper (O. Bour *et al.*, manuscript in preparation, 2002).

[9] Note finally that individualizing fractures from the mapped fracture traces are not an univocal exercise; for instance, splitting branching fracture patterns into several distinct fractures can be done in several ways. In the field, fractures and fracture tips are actually well defined from their morphologic signature. Both the continuity of fracture plane with a roughly constant orientation and the coherence of fracture aperture well define individual fractures. Direct comparison with field observations were performed to verify that small-scale maps are representative of real fracture populations. Problems arising from branching pattern do exist, but cases for which the ambiguity cannot be solved are actually few and do not affect significantly the measured statistical distributions.

#### 3.2. A First-Order Model of Fracture Population Systems

[10] The simplest geometrical model gives the number of fractures that have a length  $l$ , an orientation  $\theta$ , etc., per unit of material volume. As for fracture length or orientation, defining the fracture position is thus a first, and necessary, step in building the fracture population geometrical model. All points of fracture plane are possible a priori candidates; we take the fracture barycenter as the most “neutral” position compared to fracture tips, for instance. Physically speaking, this convenient choice lays emphasis on the nature of spatial correlations at fracture birth, since the barycenter is likely the position of the very first nuclei in the ideal case of a fracture growing symmetrically from a material point. We have verified that the definition of a fracture position is not crucial for a scaling analysis: Taking one arbitrary fracture tip, or the fracture barycenter, does not affect the scaling nature of the first-order statistical model that is described below.

[11] To describe both the spatial and length distributions of fracture networks made from sand-box models, Davy *et al.* [1990] proposed that the density fracture length distribution is adequately modeled by a double power law:

$$n(l, L) = \alpha L^{D_M} l^{-a}, \quad (1)$$

where  $n(l, L)dl$  is the number of fractures having a length between  $l$  and  $l + dl$  in a box of size  $L$ ,  $D_M$  is the mass dimension of fracture barycenters,  $a$  is the exponent of the frequency distribution of fracture lengths, and  $\alpha$  is a fracture density term. In the model, it is assumed that the length distribution of fractures follows a power law, but equation (1) is valid whatever  $D_M$ , even for a nonfractal distribution

of elements in a plane ( $D_M = 2$ ), or in a volume ( $D_M = 3$ ). The great advantage of the model presented in equation (1) is to allow the derivation of the fracture density term,  $\alpha$  which fixes the number of fractures of a given length per unit “fractal” area [Davy *et al.*, 1990]. Once validated, such a geometrical model can thus be used to derive the scaling of some properties of the fracture network, such as fracture density and fracture connectivity [Bour and Davy, 1997, 1998; Berkowitz *et al.*, 2000]. Note that equation (1) may be rewritten in the form that depends on  $N_t$ , the total number of fractures,  $l_{\min}$ , the smallest fracture detected in the system, and  $\Lambda$ , the size of the studied domain [Davy *et al.*, 1990]. However, since the latter parameters are quite difficult to measure in the field, we keep equation (1) in the form presented here, knowing that all parameters ( $\alpha$ ,  $L$ ,  $D_M$ ,  $a$ ) may be measured or derived from a fracture population.

[12] In the model the scaling of the number of elements in a box of size  $L$  comes from the definition of the mass dimension,  $D_M$ , that tells that on average, the number of elements lying in a disk or box of size  $L$  increases with  $L$  such as  $N(L) \sim L^{D_M}$  [Mandelbrot, 1982]. A simple demonstration may also be obtained through the definition of the box-counting method [Mandelbrot, 1982]. Let us consider the subsampling in a box of size  $L$  of a fractal network made of  $N_t$  elements of elementary size  $r_{\min}$ . The number  $N_b(L)$  of boxes of size  $L$  necessary to recover the fractal is given by

$$N_b(L) = (\Lambda/L)^{D_0}, \quad (2)$$

where  $D_0$  is the capacity dimension and  $\Lambda$  is the size of the box for which the fractal is completely included in one box ( $r_{\min} < L < \Lambda$ ). Thus the average number of elements included in a box of size  $L$  that contains some of the elements of the fractal is given by

$$N(L) = N_t/N_b(L) = N_t(L/\Lambda)^{D_0}. \quad (3)$$

Note that equation (3) that implies  $D_M = D_0$  is valid only if the system is monofractal (see below).

[13] To test the model (equation (1)), in sections 4 and 5 we present a statistical analysis of the geometry of the Hornelen fracture network mapped at different scales of observation (Figure 1). First, an analysis of the different maps is presented to determine the variability of the measurements from one map to another. The analysis of the fracture spatial distribution will be presented before the analysis of the length distribution because the normalization of the measurements requires the primary derivation of the mass dimension,  $D_M$ . In section 6, we present complementary developments that allow us to check the relevance of the model.

## 4. Measurements on Individual Maps

### 4.1. Fractal Dimension

[14] Different methods of measurement can be used to characterize the fractal nature of a population of elements [Feder, 1988]. Among them, the box-counting method is certainly the most well known and the most frequently used to derive the fractal dimension (here the capacity dimension  $D_0$ ) of fracture patterns [see Bonnet *et al.*, 2001]. Nevertheless, because of several bias and pitfalls, the application of this method to natural systems can be a rather troublesome

procedure [Vicsek, 1992; Walsh and Watterson, 1993; Ouilon and Sornette, 1996; Hamburger *et al.*, 1996; Bour, 1997]. Theoretically, for a fractal fracture pattern of size  $L$ , the number  $N_b(r)$  of boxes of size  $r$  should follow equation (2). However, despite the large number of fractures in each map (more than 2000), measurements made by this method on fracture centers were systematically dominated by finite-size effects (Figure 2a). For large  $r$  (i.e., larger than the largest distance between two neighbors),  $N_b(r)$  varies as  $(L/r)^2$  meaning that all boxes are occupied by one or more fracture centers. This trivial scaling occurs for  $r > 8$  m in the example shown in Figure 2a ( $L = 55$  m), and the dimension derived is that of the sampling domain instead of the dimension of the fracture network [Walsh and Watterson, 1993]. Another sampling effect occurs at very small  $r$  ( $< 0.3$  m in Figure 2a), when most occupied boxes cover only one fracture center which leads to  $N_b(r)$  being equal simply to the number of fractures in the system,  $N$  (Figure 2a). The question now is to know if a dimension can be calculated in the intermediate scale range, that is, between  $\sim 0.3$  and 8 m in the example shown in Figure 2a. The answer is intimately related to the modeling of finite-size effects. To illustrate this point, we draw in Figure 2a a simple “sampling” function that has the expected behavior at large and small  $r$ :

$$N'_b = \frac{1}{N^{-1} + (r/L)^2}.$$

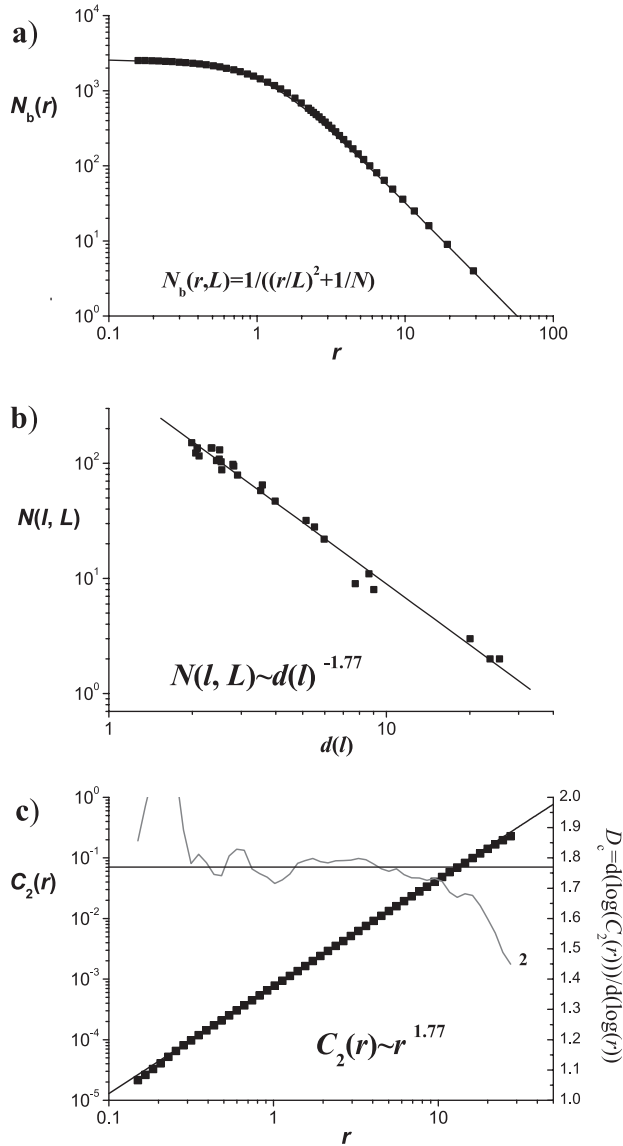
This function appears to efficiently model data over the whole range of scale, showing that sampling effects may remain dominant at all scales, prohibiting the derivation of a fractal dimension by this method. However, the fact that  $N_b(r)$  varies as  $(L/r)^2$  at large scale does not statistically mean that the capacity dimension of the fracture network is equal to 2 in regard to the small range of scales for which this relationship is valid.

[15] We propose a modified method to derive the capacity dimension  $D_0$ , which consists of calculating the average distance  $d(l)$  between nearest centers of fractures having the same length. This method, which was initially proposed by Bour and Davy [1999], assumes that any subset of fracture centers remains fractal. The idea is that the number of disks necessary to cover the fractal will be equal to the number of sampled points if the disk radius is about the average of the distances between nearest points. Following equation (2), the number of sampled points  $N$  should scale with the distance between nearest points  $d$  as

$$N \sim (L/d)^{D_0}. \quad (4)$$

We can select the subset of fractures at random or with respect to their length. We use the latter selection because it additionally provides helpful information to verify equation (1). Note that no assumptions are required for the length distribution. If the length distribution is a power law, equations (1) and (4) predict  $d(l)$  to vary as  $d(l) \sim l^x$ , with  $x = (a - 1)/D_0$ .

[16] We have thus calculated  $N(l, L) = n(l, L)dl$ , which is the number of fractures having a length in the range  $[l, l + dl]$  and contained in a system having a typical size  $L$ , and  $d(l)$  the average distance between fractures of same length  $l$ . Figure 2b



**Figure 2.** Analysis of the spatial distribution of the fracture pattern having a size  $L$  equal to 55 m. (a) Box-counting method: plot of the number of boxes necessary to cover the system,  $N_b(r)$  versus  $r$ , the size of the boxes. Solid squares correspond to calculations, while the solid curve corresponds to the function  $f(r) = 1/[(r/L)^2 + 1/N]$ , where  $N$  is the total number of fractures included in the system. (b) Plot of the number of fractures having a length between  $l$  and  $l + dl$  (here a logarithmic bin is chosen meaning that  $dl$  is proportional to  $l$ ),  $N(l, L)$ , versus the average distance between these fractures,  $d(l)$ .  $N(l, L)$  is fitted by  $N(l, L) \approx \beta d(l)^{-D_0}$  (solid straight line) with  $D_0 = 1.77 \pm 0.08$ . Most of the uncertainty comes from the choice of the scale range chosen to perform the fit. (c) Computations of the two-point correlation function,  $C_2(r)$ , as a function of  $r$ .  $C_2(r)$  is fitted by  $C_2(r) \approx \alpha r^{-D_c}$  (solid straight line) with  $D_c = 1.77 \pm 0.02$ . The uncertainty may also be visually estimated from the curve labeled 2, which gives the local value of the logarithmic slope  $d[\log(C_2(r))]/d[\log(r)]$  (right-hand scale).

shows that equation (4) is verified with  $D_0 \sim 1.77 \pm 0.08$ . As in any scaling analysis, the uncertainty estimate on the fractal

dimensions depends on the data scattering, on the choice of a fitting procedure (maximum likelihood, least square, regression, etc.), and on the scale range used to performed the fit. The latter point is often neglected in spite of the fact that this is the major cause of uncertainty. The values that are reported through the text are calculated by considering that the upper and lower bounds of the fitting model can vary.

[17] Apparently, the procedure used to derive the capacity dimension  $D_0$  seems to not have the drawbacks of the box-counting method and provides a simple but crude estimate of  $D_0$ , knowing that its precise determination remains anyway a difficult challenge for natural systems [Vicsek, 1992; Walsh and Watterson, 1993; Ouillon and Sornette, 1996; Hamburger et al., 1996; Bonnet et al., 2001]. Note that the determination of the correlation dimension  $D_2$ , which is presented below, is much more accurate.

[18] To compute the fractal dimension of a natural system, the most efficient method according to Vicsek [1992] consists of calculating the so-called density-density or pair correlation function  $C_2(r)$ , defined as

$$C_2(r) = 2N_p(r)/[N(N-1)],$$

where  $N_p(r)$  is the number of pairs of points whose distance is less than  $r$  [Hentschel and Proccacia, 1983] and  $N$  is the total number of points.  $C_2(r)$  is expected to scale with  $r$  such as  $r^{D_c}$ , where  $D_c$  is the correlation dimension. This method has been implemented on fracture barycenters [Davy et al., 1990; Sornette et al., 1993; Bour and Davy, 1999]. Theoretically, the correlation dimension  $D_c$  is identical to the mass dimension  $D_M$ ; the two-point correlation function therefore provides an accurate method to derive the mass dimension that fixes the scaling of the number of fractures in a box of size  $L$  in the model presented in equation (1). We indeed verified that both methods lead to similar results [Bour, 1997]. Note nevertheless that to obtain  $D_c = D_M$  on the same scale range, resolution effects must be partly removed from the computations of the mass dimension. In particular, the computations of the number of elements lying in a disk of radius  $r$  should not include the fracture midpoint over which the disk is centered [Bour, 1997]. An example of the computation of the two-point correlation function is given in Figure 2c where the quality of the power law fit may be appreciated from the local logarithmic slope also drawn on Figure 2c ( $d[\log(C_2(r))]/d[\log(r)]$ ) which is equal to  $D_c$  if  $C_2$  is a power law). The power law fit is valid over almost two decades with a dimension  $D_c$  of  $\sim 1.77$  whatever the regression method. The uncertainty is  $< 0.02$ , emphasizing the accuracy of this correlation method. It is mainly due to the choice of scale range on which the regression is applied.

[19] In a monofractal object the fractal dimension fully characterizes the variation with scale of the object density, and of its distribution moments  $M_q(r) \sim r^{(q-1)D}$ . In more complex objects a single exponent is not sufficient to fully describe the scaling of the moments of the density distributions. These objects are defined as multifractal and are characterized by a set of generalized dimensions  $D_0, D_1, \dots, D_q$  that constitutes a decreasing series. The correlation dimension  $D_c$  is equal to  $D_2$ , and the correlation integrals of triplets, quadruplets,  $\dots$ ,  $n$ -tuplets provide  $D_3, D_4$ , etc. [Hentschel and Proccacia, 1983]. In the example shown in Figure 2 the generalized dimensions are found to be about

**Table 2.** Values of the Measured Parameters for the Different Fracture Networks<sup>a</sup>

| $L$ | $D_0$           | $D_c$           | $D_3$           | $D_4$           | $a$  | $\alpha$ | $\alpha'$ |
|-----|-----------------|-----------------|-----------------|-----------------|------|----------|-----------|
| 18  | $1.80 \pm 0.1$  | $1.78 \pm 0.02$ | $1.76 \pm 0.03$ | $1.76 \pm 0.05$ | 2.73 | 3.94     | 3.72      |
| 55  | $1.77 \pm 0.08$ | $1.77 \pm 0.02$ | $1.75 \pm 0.03$ | $1.74 \pm 0.03$ | 2.66 | 3.73     | 3.31      |
| 90  | $1.80 \pm 0.05$ | $1.77 \pm 0.02$ | $1.74 \pm 0.03$ | $1.73 \pm 0.05$ | 2.53 | 2.80     | 2.45      |
| 90  | $1.80 \pm 0.1$  | $1.80 \pm 0.04$ | $1.79 \pm 0.04$ | $1.79 \pm 0.05$ | 2.76 | 4.50     | 4.50      |
| 180 | $1.82 \pm 0.1$  | $1.82 \pm 0.04$ | $1.80 \pm 0.03$ | $1.79 \pm 0.03$ | 2.74 | 3.75     | 4.16      |
| 360 | $1.85 \pm 0.1$  | $1.82 \pm 0.05$ | $1.82 \pm 0.05$ | $1.82 \pm 0.05$ | 3.00 | 8.27     | 9.30      |
| 720 | $1.88 \pm 0.1$  | $1.87 \pm 0.1$  | $1.87 \pm 0.10$ | $1.84 \pm 0.10$ | 2.85 | 3.96     | 6.28      |

<sup>a</sup> $L$ , size of the system (meters);  $D_0$ , capacity dimension;  $D_c$ , two-point correlation dimension;  $D_3$ , three-point correlation dimension;  $D_4$ , four-point correlation dimension;  $a$ , exponent of the power law length distribution;  $\alpha$  fracture density term (calculated through equation (1) and using the table of values of  $D_c$ );  $\alpha'$  fracture density term (calculated through equation (1) with  $D_c = 1.8$  for all fracture patterns). Uncertainties are mainly due to the choice of the scale range used to perform fit.

equal (see also Table 2,  $L = 55$  m), emphasizing that fracture barycenters seem to be monofractal. A more detailed discussion about the mono or multifractal nature of the Hornelen fracture network will be given in section 4.3.

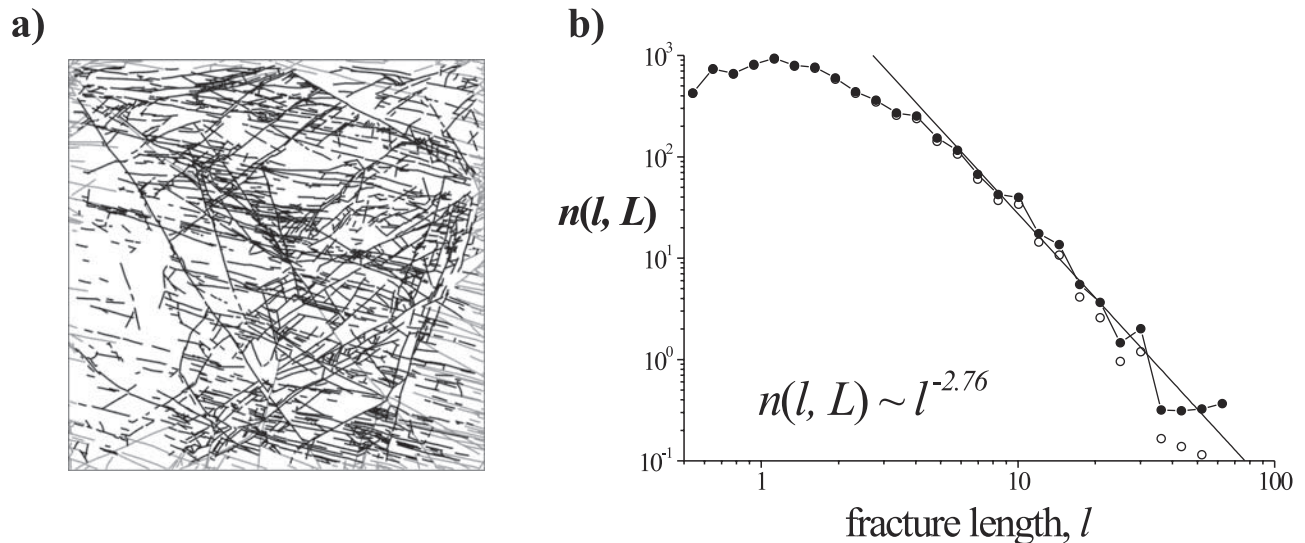
[20] Note that the previous results are valid whatever the point used to define fracture location (barycenter, fracture tips or any point taken at random in the fracture line) [Bour, 1997]. The reason comes from the fact that the derivation of two-point correlation dimension is statistically dominated by the numerous small fractures, for which the error in the determination of the precise spatial location is relatively insignificant. For the same reason, we also found that censoring effects (fractures that intersect the system boundaries), that have to be explicitly taken into account when deriving the length distribution model [Pickering *et al.*, 1995], did not affect the derivation of the fractal dimension.

#### 4.2. Length Distribution

[21] The main problem arising when assessing a model of length distribution comes from truncating and censoring

effects [Laslett, 1982; Pickering *et al.*, 1995; Clark *et al.*, 1999]. Truncating effects result from the scale of resolution which limits the observation of fractures to the ones that are greater than a given length. Below this resolution length, the proportion of fractures not detected depends on their length in some unknown function. As a consequence, any power law model of length distribution will be limited by a lower cutoff.

[22] A few procedures have been proposed to correct the measurements for censoring effects [Pickering *et al.*, 1995; Ouillon *et al.*, 1996b; Clark *et al.*, 1999]. Here we used a procedure derived from a previous work by Laslett [1982], which consists of estimating the probability of occurrence of a fracture of length  $l$  depending on the size of the window of observation,  $L$  [Bour and Davy, 1999]. The data corrected for finite-size effects can be fitted by a power law over  $\sim 1 - 1.5$  decades with an exponent of  $\sim 2.8$  in the example shown in Figure 3. The uncertainty in the exponent is strongly dependent on the scale range chosen to perform the fit. We estimate the uncertainty to be  $\sim 0.3 - 0.5$  depending on the number of fractures.



**Figure 3.** Density length distribution of the fracture pattern having a size  $L = 90$  m. (a) Fracture pattern. Fractures which intersect the boundaries and whose real lengths are not known are not taken into account in the calculations of the density length distribution (such fractures are represented in gray). By doing so, the corrections for censoring effects are simplified since the probability of a fracture to lie entirely in the system depends mainly on its length and on the system size [see Bour and Davy, 1999]. (b) Density length distribution. Measurements corrected from censoring effects are in solid symbols while noncorrected measurements are in open symbols. Because of the truncation effects, the length distribution follows a power law only for length  $> 5$  m. The solid straight line corresponds to a power law regression for  $l$  ranging between 5 and 65 m with  $a = 2.76$ .

### 4.3. Comparison Between Maps

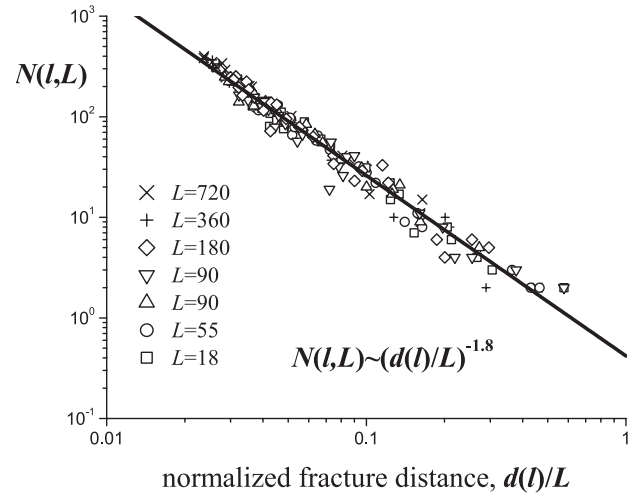
[23] Table 2 summarizes the results obtained from the different maps of the Hornelen basin. We found very similar results for the different exponents that characterize the fracture networks. For the five smallest maps ( $L$  ranging from 18 to 180 m) the generalized dimensions are similar, with a dimension of  $\sim 1.8$ . No significant variation with system size is detectable. With respect to the calculated uncertainties, it may be concluded that the system is monofractal since all generalized dimensions are about equal. However, taking into account the difficulty of deriving precisely the capacity dimension  $D_0$ , we cannot exclude the possibility of a slight multifractal component in the fracture scaling pattern.

[24] For the largest map ( $L = 720$  m) the dimensions  $D_q$  are larger than for other maps. However, even the correlation dimensions are not very well defined, with a local logarithmic slope that systematically decreases with increasing  $r$  (see Figure 5c in section 5). This systematic decrease makes the determination of fractal dimensions extremely dependent on the choice of a scale range. At small scales, that is, between 8 and 30 m, the apparent dimension  $D_2$  is  $\sim 2$ ; between 30 and 200 m, it is  $\sim 1.85$ . Note that we cannot invoke a physical change of the fracturing regime to explain this variation since the small-scale value is inconsistent with the one measured on other maps that also sample the same scale range (see Figure 5a in section 5). A possible explanation is that the lines drawn from this large-scale map may represent fracture zones and not individual fractures. The correlation measured by the fractal dimension may thus have a different physical meaning than for other maps. Another possible explanation is that the sampling procedure itself may alter correlation at small scales (C. Darcel et al., manuscript in preparation, 2001). The fact that such an effect (a higher dimension at small scale) can be observed in all maps derived from aerial photographs, in a lesser measure, however, seems to corroborate this statistical explanation.

[25] The exponent of the power law length distribution,  $\alpha$ , varies from 2.5 up to 3.0 depending on the fracture network. With exponents  $\alpha$  and  $D_c$  measured, it is possible to deduce the density term  $\alpha$  from equation (1) for each fracture map. We find  $\alpha$  varying from 2.8 to 8.3, which corresponds to a relatively low variability knowing that a small variation of  $\alpha$  leads to a large variation of  $\alpha$  (Table 2). Since close values of  $\alpha$ ,  $a$ , and  $D_c$ , are found for all fracture maps, it seems reasonable to look for a unique distribution law, such as in equation (1), for the whole fracture network. This issue is addressed in section 5, in which we discuss the methods for normalizing the different data sets.

### 5. Scale Normalization

[26] The question we address in this section is how to combine the measurements made on the different maps to check the relevance of a global statistical model for all fracture maps. We discuss the scale dependency of the measured parameters and the correct normalization that makes possible comparison between maps. Problems arise for parameters that depend on the resolution scale of maps because the scale varies from map to map. For instance, the total number of fractures, the number of fractures smaller than a certain length, or the average distance between



**Figure 4.** Computations of  $N(l, L)$  as a function of the ratio  $d(l)/L$  for all fracture patterns.  $N(l, L)$  is fitted for all fracture patterns by  $N(l, L) \approx (d(l)/L)^{-D_0}$  (thick solid straight line) with  $D_0 = 1.8 \pm 0.1$ .

fractures is dependent on the resolution scale. Other parameters, like the cumulative number of fractures larger than a given length, depend only on the system size. For these parameters the scaling with the system size  $L$  should be related to the statistical model of fracture distribution. Equation (4) is an example of such scaling normalization that relates the average distance between fractures to the system size and to the number of fractures. In sections 5.1 and 5.2,  $\epsilon$  refers to the resolution scale of maps, and  $L$  is the typical length scale of the sampled domain.

#### 5.1. Fractal Dimension

[27] According to equation (4) the number of fractures having a length between  $l$  and  $l + dl$ , that is,  $N(l, L)$  shown in Figure 2b, should only depend on the ratio  $d(l)/L$ . In Figure 4 we plot  $N(l, L)$  as a function of the normalized fracture distance  $d(l)/L$ . The data collected from all maps string together along a single power law of exponent  $-1.8 \pm 0.1$ . The fit is good even for fractures clearly below the resolution scale of maps for which the number is underestimated. The reason being that if equation (4) is valid for a network, it remains valid for any subset as long as the subset remains fractal. In that case, the underestimation of the element number is compensated for by an increase of the average distance in a way that follows equation (4). Note that an implicit condition for a subset to remain fractal is that the undersampling has no systematic bias (i.e., is made at random). We assume that this condition was fulfilled when mapping fracture patterns. In the following paragraph, we propose a normalization of the two-point correlation function that allows a much more precise scaling comparison between maps.

[28] The correlation integral measure is based on the number of pairs of elements,  $N_p(r)$ , whose distance is less than  $r$ . This measure is clearly dependent on the total number of sampled points, and thus on the resolution scale  $\epsilon$  of the map and on the system size  $L$ . In contrast, the function  $C_2(r/L) = 2N_p(r/L)/[N(N-1)]$  is independent of both  $\epsilon$  and  $L$  at least in the range of scales where fractures

are correctly sampled, that is, for  $r > \varepsilon$ . This assessment has been verified numerically by varying both  $\varepsilon$  and  $L$  on one fracture map. Changing  $\varepsilon$  affects  $C_2$  only for  $r < \varepsilon$ , and the normalization by  $N(N-1)$  and  $r/L$  makes  $C_2$  independent of the system size (this is obvious for  $r = L$  since  $C_2(1) = 1$  whatever  $L$ ). If  $D_c$  is identical for all maps, the measures  $C_2(r/L)$  should scale with a single power law with an exponent  $D_c$ . In Figure 5b, we compare  $C_2(r/L)$  for the different fracture maps. We observe that the different curves are mapped onto a single curve and are thus indistinguishable if  $r > L/20$ . Even finite-size effects, when  $r$  goes to  $L$ , are similar. For a scale range corresponding to  $L/20 < r < L/5$ , the best fit for all fracture maps gives  $D_c$  of  $\sim 1.8$  (Figure 5b) with an accuracy of  $\sim 0.04$ . The accuracy of the fit may be also appreciated from the local logarithmic slope drawn in the Figure 5c. For the four smallest maps ( $L = 18$  to  $L = 90$ ) the fit even appears valid from  $L/200$  to  $L/5$ , which represents more than 1 order of magnitude for each map and almost 3 orders of magnitude if the four maps are combined.

[29] From the previous results it seems that a representative value of  $D_c$  of  $\sim 1.8$  can be reasonably used to describe the fractal properties of the Hornelen fracture network. Despite a larger uncertainty, we also found  $D_0$  of  $\sim 1.8$  (Figure 4) as with most of all the generalized dimensions (Table 2). For notation simplicity the fractal dimension  $D$  refers to the correlation dimension  $D_c = D_2$  in the following, knowing that the all other fractal dimensions are close or about equal to  $D$ .

## 5.2. Length Distributions

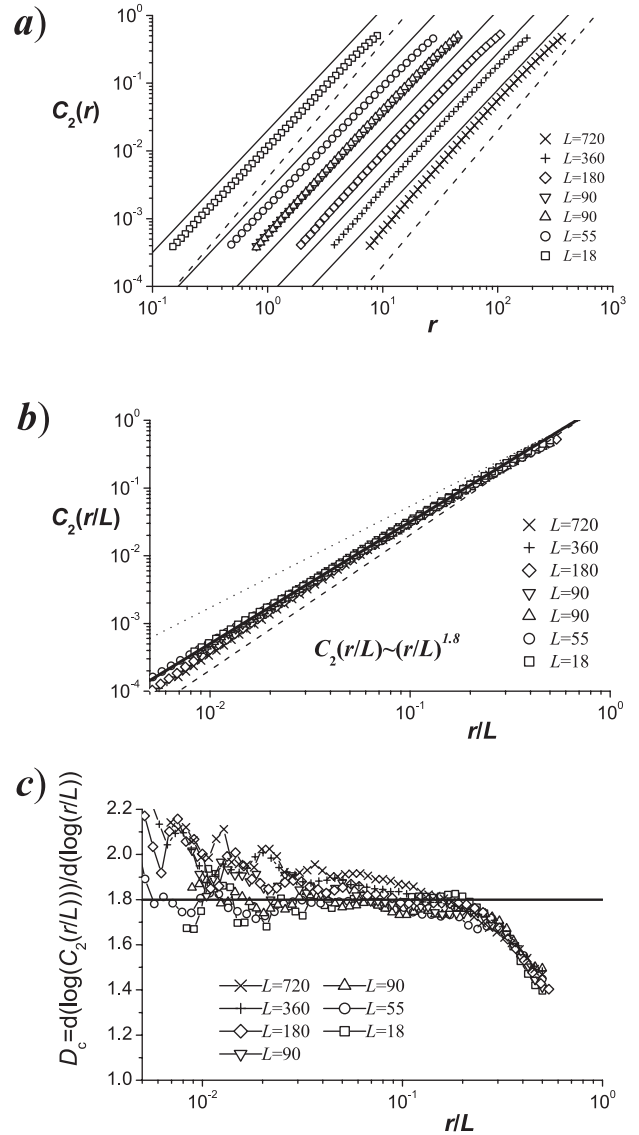
[30] The normalization of the length distribution is necessarily dependent on the density model that fixes the relationship between the number of fractures and the system size  $L$ . A common normalization model calculates the number of fractures per unit area [Heffer and Bevan, 1990; Yielding et al., 1992; Scholtz et al., 1993; Castaing et al., 1996; Odling, 1997]. This implicitly assumes that the density is constant whatever the observation scale. In the preceding, we have shown that this assumption is not correct for the Hornelen basin and that the density model has a fractal property with a dimension  $D$  of  $\sim 1.8$ . The different length distributions should then be normalized according to equation (1). In Figure 6 we plot  $n(l, L)/L^D$  as a function of the fracture length  $l$ . Except for lengths below the resolution scales, data all map onto a single power law characterized by an exponent  $-a$ , with  $a \sim 2.8$ . This scaling law is valid over more than two decades. The fractal density term ( $\alpha$  in equation (1)) is equal to  $\sim 4.5$ . According to the different measurements it seems that the model presented in equation (1) represents adequately the length and spatial distribution of all fracture networks with only two basic exponents and one density term.

## 6. Other Validations of the Statistical Model

### 6.1. Distance Between Fractures

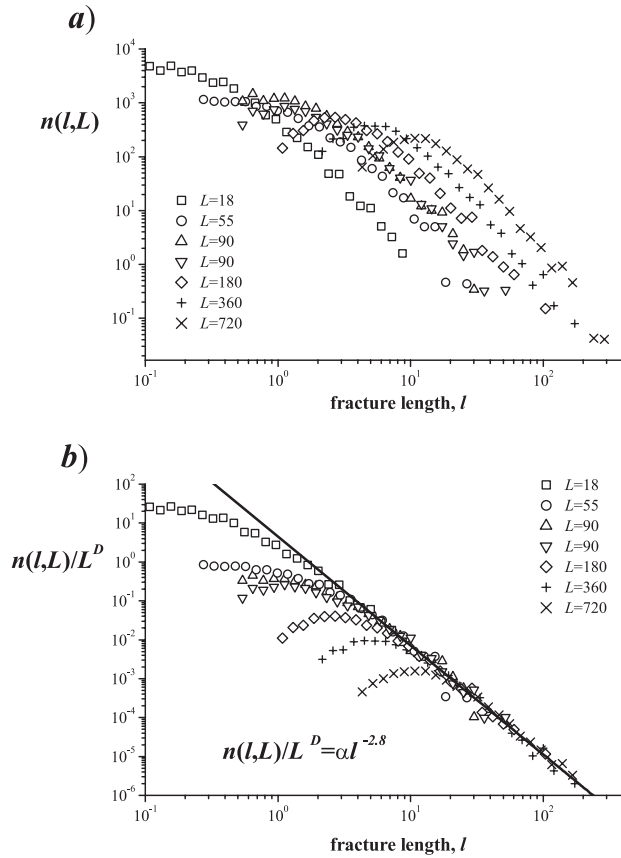
[31] We have shown that the average distance between nearest fracture centers of the same length fracture (i.e., in the interval  $[l, l + dl]$ ) should scale as  $l^x$ , where  $l$  is the fracture length and  $x$ , an exponent relating the length exponent  $a$  and the fractal dimension  $D_0$  by

$$x = (a - 1)/D_0. \quad (5)$$



**Figure 5.** Computations of the two-point correlation functions for all fracture maps. (a) Computations of the non normalized two-point correlation function,  $C_2(r)$ , as a function of  $r$ . Solid and dashed lines correspond to  $C_2(r) \approx (r)^{1.8}$  and  $C_2(r) \approx (r)^{2.0}$ , respectively. (b) Computations of the normalized two-point correlation functions  $C_2(r/L)$  as a function of  $r/L$ .  $C_2(r/L)$  is fitted for all fracture patterns by  $C_2(r/L) \approx (r/L)^{D_c}$  (thick solid straight line) with  $D_c = 1.8$ . Dashed and dotted line correspond to  $C_2(r/L) \approx (r/L)^2$  and  $C_2(r/L) \approx (r/L)^{1.5}$ , respectively. (c) Variations of the apparent logarithmic slope  $d[\log C_2(r/L)]/d[\log(r/L)]$  for the different fracture patterns.

The average distance between fractures having the same length,  $d(l)$ , may be easily calculated for the different maps. It is independent of the scale resolution and of the system size, as long as all fractures of that length are sampled. A direct comparison of the curves  $d(l)$  for the different maps is thus meaningful. We present the results in Figure 7 which shows that  $d$  scales about linearly with  $l$  if we restrict the analysis to the lengths larger than the map resolution scales.



**Figure 6.** Density length distributions measured for the different fracture patterns. (a) Nonnormalized density length distributions. (b) Normalized density length distribution. The solid straight line corresponds to a power law regression for  $l$  ranging between 2 and 200 m with  $a = 2.8$ .

A value slightly larger or smaller than  $x = 1$  is inconsistent with the whole set of measurements (dashed and dotted lines, Figure 7);  $x = 1$  is consistent with  $a = 2.8$  and  $D_0 = 1.8$ , the values derived from independent analyses, and implies the self-similarity of the Hornelen fracture network (see also section 7).

## 6.2. Limited Observation of Fracture Lengths

[32] To complete this analysis, we propose a procedure which is both a validation test of the power law model through scale and a way to derive the exponent  $a$  and the parameter  $\alpha$  even when only a small part of the length distribution is correctly represented. The derivation of the exponent  $a$  is indeed a tricky exercise when dealing with scarce natural data. We know that censoring and truncation effects limits the range of length on which a power law fit can be applied [Pickering et al., 1995; Clark et al., 1999], but generally, we do not know how much this limitation affects the derived exponent. The approach proposed here assumes that the density of fractures is correctly measured for only a given fracture length  $l_c$ . Given the difficulty of mapping detailed fracture networks, this corresponds to most of the sampling conditions for which the fracture

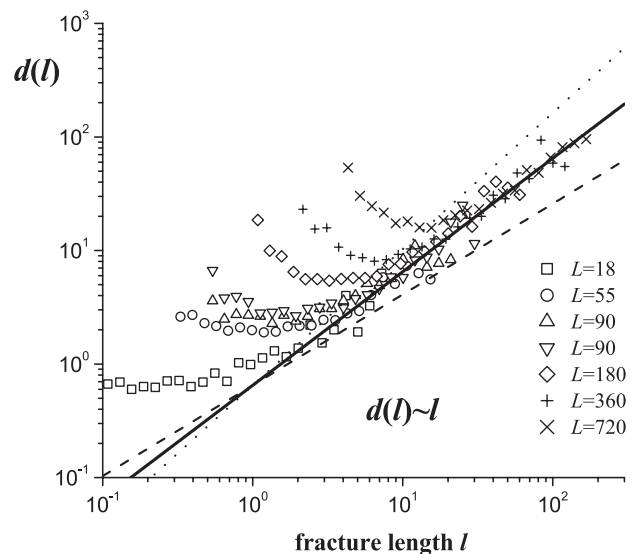
length distribution is systematically curved when plotted on a bilogarithmic graph due to large censoring and truncation effects, making a reliable determination of the power law length exponent impossible.

[33] If the fracture set is complete for fractures of length around  $l_c$ , it is possible to calculate the fracture length density  $n(l_c, L) = N(l_c)/dl$ , where  $N(l_c)$  is the number of observed fractures whose length is in the range  $[l_c, l_c + dl]$ . If the equation (1) is valid,  $a$  and  $\alpha$  are related to  $n(l_c, L)$  by

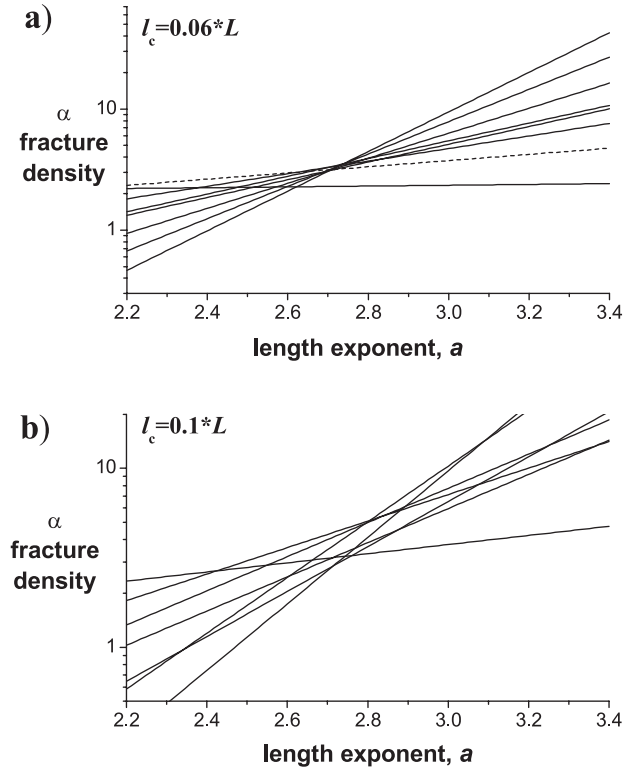
$$\log(\alpha) = a \log(l_c) + \log(n(l_c, L)/L^D). \quad (6)$$

In the space  $(a, \log(\alpha))$ , equation (6) is the equation of a straight line whose slope is equal to  $\log(l_c)$ . The straight line gives the ensemble of solutions compatible with the measure  $n(l_c, L)$ . With different maps at different scales, we restrict the range of possible values of  $a$  and  $\alpha$ . In practice, we draw different straight lines corresponding to (6) for different maps, these should all intersect in one point of the domain  $(a, \log(\alpha))$  if the statistical model is valid through all scales.

[34] We apply the method to the Hornelen data set for different values of  $l_c$  (Figure 8). Assuming  $D = 1.8$  and  $l_c$  systematically equal to 6% of the system sizes, all straight lines except one intersect on a unique point, defining precisely  $a = 2.75 \pm 0.05$  and  $\alpha = 3.5 \pm 0.5$ . The method gives a much better accuracy than the normalized length distribution shown in Figure 6. The only curve that does not intersect the other ones is for the smallest system with  $L = 18$  m for which the ratio  $n(l_c, L)/L^D$  is relatively small. Odling [1997] argues that 1 m, that is, the value chosen for  $l_c$ , is a possible physical limit to the power law behavior for



**Figure 7.** Distance distributions measured for the different fracture patterns. For each fracture pattern the average distance between fractures having the same length,  $d(l)$  is calculated and plotted versus the fracture length  $l$ . The solid straight line corresponds to a power law model so that  $d(l) \sim l^x$  with  $x = 1$ . The dashed and dotted lines correspond to power law models with  $x = 0.8$  and  $x = 1.2$ .



**Figure 8.** Plot of equation (6) for all fracture patterns using different values of  $l_c$ : (a)  $l_c = 0.06L$  and (b)  $l_c = 0.1L$ . The dashed line in Figure 8a corresponds to the smallest map ( $L = 18$  m) with  $l_c = 2$  m. The higher is the slope of the straight line, the larger is the size  $L$  of the map.

this fracture system. Reassigning  $l_c$  equal to 2 m (dashed line in Figure 8) gives a line that is perfectly consistent with the others.

[35] The choice of the relevant scale  $l_c$  remains the main difficulty in applying this method. In Figure 8b we show a similar analysis with a larger value  $l_c$  of 10% of the system size. The straight lines do not intersect in a single point as in Figure 8a, but on a larger domain. The uncertainty on  $a$  is  $\sim 0.3$  resulting from the small number of fractures and from possible censoring effects. Similar poor results are obtained if  $l_c$  is too small. For instance, if  $l_c = 2\%$  of the system size, the uncertainty on  $a$  is still  $\sim 0.3$  and is much larger than for  $l_c = 6\%$  of the system size. Thus a trial-and-error approach may be used to determine the optimal values of  $l_c$ . This method validates the assumption that a unique statistical model may describe the fracture networks at all scales because all curves intersect at a single value of  $a$  and  $\alpha$ .

## 7. Discussion and Conclusions

[36] We have proposed and tested a first-order model of the geometry of natural fracture networks that provides the average number of fractures of a given length at a given scale. This model involves three parameters: the exponent  $a$  of a power law length distribution, the fractal dimension  $D$  which fixes the scale dependence of the number of fractures, and the fracture density  $\alpha$ . We verify the applicability of this

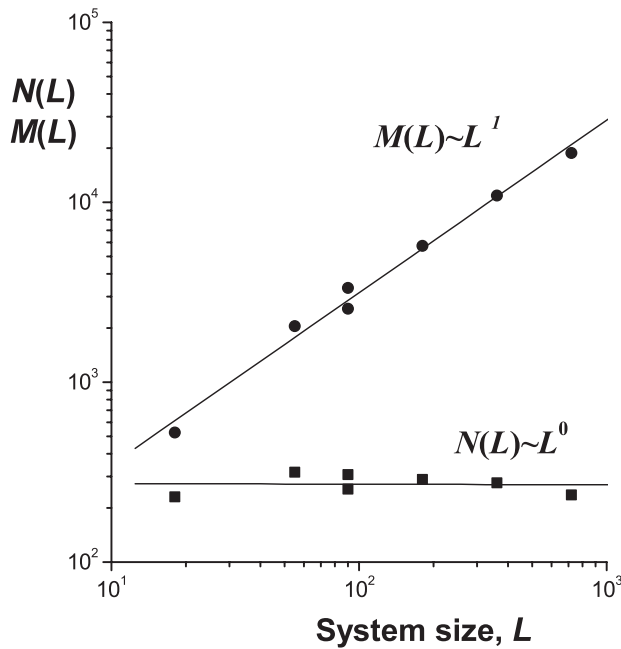
model on seven fracture patterns mapped from the metric scale up to almost the kilometric scale in the Hornelen basin. The model is found to efficiently describe the fracture network properties at all scales with a single set of parameters,  $a = 2.75$ ,  $D = 1.8$ , and  $\alpha = 3.5$ .

[37] The previous analysis of the data from Hornelen [Odling, 1997] normalized the length distribution with respect to the area of the sample  $L^2$  (equivalent to assuming a fractal dimension of 2.0) and used the cumulative rather than the density length distributions. The individual length distributions were also corrected for censoring effects using a different method (the Kaplan-Meier method [see Odling, 1997]). Although the normalization scheme used by Odling [1997] did not take into account the fractal nature of the system, this initial treatment appears to be internally consistent as the corrected normalized distributions lie along the same trend. The exponent derived from this initial analysis is equivalent to an exponent  $a$  of 3.1. This compares fairly well with the derived exponent  $a$  of 2.75 from the analysis in the present paper. The difference in values of  $a$  are mainly a result of the different normalization schemes used. Although these two exponents are fairly close, they could, if used to predict behavior over a wide range of scales, lead to different conclusions about fracture density and fracture connectivity. These two approaches to analyzing the data elegantly illustrate the difficulty in quantifying scaling behavior of natural systems which requires an analysis of both the length and spatial distributions.

[38] Once validated, such a geometrical model can be used to derive the scaling of some properties of the fracture network, such as fracture density and fracture connectivity [Bour and Davy, 1997, 1998; Berkowitz et al., 2000]. For instance, one can predict from equation (1) the evolution with scale of the number of fractures,  $N(L)$ , or of the cumulative length of fractures,  $M(L)$  included in a system of size  $L$ . Indeed,  $N(L)$  is given by

$$N(L) = \int_{l_{\min}}^{l_{\max}} n(l, L) dl = \alpha \frac{L^D (l_{\min}^{-a+1} - l_{\max}^{-a+1})}{a-1}, \quad (7)$$

where  $l_{\min}$  and  $l_{\max}$  are the smallest and the largest fracture, respectively, contained in the system. For  $a > 1$  the upper bound of the integral can be neglected so that  $N(L)$  only depends on  $L$  and  $l_{\min}$ . Similar scaling have been determined experimentally by Castaing et al. [1996] and by Odling [1997], who measured the variations with the system size or with the observation height of the apparent fracture density per unit area. For instance, using the results of Odling [1997], we obtained a scaling for the normalized apparent density ( $M(L)/L^2$ ) such as  $M(L)/L^2 \sim L^{-0.86}$ . This result is close to the one expected from equation (8) with  $a = D + 1$  which predicts  $M(L)/L^2 \sim L^{-1}$ , when  $l_{\min}$  is fixed to a given percentage of the system size. To verify the latter result on the different fracture maps, one must take care of the non-trivial variation of  $l_{\min}$  with  $L$ , the sample size, and of truncation effects which limit the range of length over which equation (1) is valid. For these reasons, we compute  $N(L)$  and  $M(L)$  only for fractures having a length larger than a length  $l_c$  fixed to a certain percentage of the map size. Here, we fixed  $l_c = 6\%$  in agreement with results obtained



**Figure 9.** Scaling with  $L$  of  $N(L)$  (squares) and of  $M(L)$  (circles).  $N(L)$  is the total number of fractures included in the system of size  $L$  and having a length larger than  $l_c = L/15$ .  $M(L)$  is the total cumulative length of fractures having a length larger than  $l_c = L/15$  which are included in the system of size  $L$ .

previously. Equation (7) then predicts that  $N(L)$  and  $M(L)$  vary with  $L$  as  $L^{-a+D+1}$  and  $L^{-a+D+2}$ , respectively. Indeed, we obtained  $N(L) \sim L^0$  and  $M(L) \sim L^1$  in agreement with derived values of  $a$  and  $D$  (Figure 9).

[39] The fact that  $a \approx D + 1$  implies that the fracture network of the Hornelen basin is almost self-similar. Self-similar systems are defined as systems which are statistically similar at all scales so that a part of the system will look the same as the whole system once rescaled in length. For fracture networks following a large length distribution, it necessarily implies that both the spatial and length distributions are following scaling laws but also that the exponents of these scaling laws should be related through  $a = D + 1$ . For  $a \neq D + 1$  the number of elements larger than  $\alpha L$ , with  $\alpha$  an arbitrary constant, depends on the system size  $L$  (equation (7)) so that the system cannot look the same at all scales. Such a relationship ( $a = D + 1$ ) is verified for our fracture data sets but also for the fragmentation model of *Turcotte* [1986].

[40] From a methodological point of view we present different methods for deriving the basic exponents  $a$  and  $D$  and for verifying the consistency of the model. Note that despite the large number of fractures in each map (more than 2000), the box-counting method is found to be so sensitive to finite-size effects that the fractal dimension cannot be derived through this method. The two-point correlation function appears much more suitable for demonstrating the presence of fractal scaling for fracture patterns certainly because of the stochastic nature of such objects [Vicsek, 1992]. The main methodological issue concerns the normalization of statistical measures to make

possible the intercomparison between maps at different scales. The fractal dimension  $D_0$  can be derived from the relationship between the number of fractures of a given length, and the average of the distances between nearest fracture centers (Figure 4). The normalized correlation integral  $C_2(r/L)$  gives an accurate determination of the correlation dimension (Figure 5). The length distribution normalized by the fractal density  $L^D$  gives a power law fit over several decades (Figure 6). Finally, we proposed a new method to test the consistency of the statistical model over scales and to determine accurately the length exponent  $a$  and the density term  $\alpha$  (Figure 8).

[41] To conclude, it should be pointed out that the model presented here is a first-order model that provides only the characterization of the number of fractures of a given length and at a given scale. This model can be also extended to predict the number of fractures of a given throw (or aperture) at a given scale if the throw (or aperture) distribution is known. The application of the model to one- or three-dimensional systems is straightforward. It can also be easily improved to incorporate the orientation distribution. However, possible correlation between the different geometrical parameters are not described through this model. An extended model which takes account of the observed correlation between local fracture density and fracture length [Ackermann and Schlische, 1997; Bour and Davy, 1999] will be presented in a future work.

## Notation

|           |                                                                       |
|-----------|-----------------------------------------------------------------------|
| $l$       | fracture length.                                                      |
| $n(l, L)$ | density length distribution, equal to $N(l, L)/dl$ .                  |
| $N(l, L)$ | frequency length distribution.                                        |
| $a$       | exponent of the density length distribution.                          |
| $D$       | fractal dimension.                                                    |
| $D_0$     | capacity dimension.                                                   |
| $D_M$     | mass dimension.                                                       |
| $D_c$     | correlation dimension.                                                |
| $d(l)$    | average distance between centers of fractures having the same length. |
| $L$       | size of a subsystem of the fracture network.                          |
| $N$       | total number of fractures in the whole system of size $\Lambda$ .     |
| $x$       | exponent of the distance distribution $d(l)$ .                        |
| $\alpha$  | scale-invariant fracture density.                                     |
| $\Lambda$ | size of the whole system embedding the fracture network.              |

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